what are the differences between SIMULATION and SYNTHESIS

Simulation <= verify your design. synthesis <= Check for your timing Simulation is used to verify the functionality of the circuit..
a)Functional Simulation:study of ckt's operation independent of timing parameters and gate delays.

b) Timing Simulation :study including estimated delays, verify setup,hold and other timing requirements of devices like flip flops are met.

Synthesis:One of the foremost in back end steps where by synthesizing is nothing but converting VHDL or VERILOG description to a set of primitives(equations as in CPLD) or components(as in FPGA'S)to fit into the target technology.Basically the synthesis tools convert the design description into equations or components

What are the differences between latches & flipflops?

There are 2 types of circuits:

1. Combinational

2. Sequential

Latches and flipflops both come under the category of "sequential circuits".

Difference: Latches are level-sensitive, whereas, FF are edge sensitive. By edge sensitive, I mean O/p changes only when there is a clock transition.(from 1 to 0, or from 0 to 1)

Example: In a flipflop, inputs have arrived on the input lines at time= 2 seconds. But, output won't change immediately. At time = 3 seconds, clock transition takes place. After that, O/P will change.

Flip-flops are of 2 types:

1.Positive edge triggered

2. negative edge triggered

1)fllipflops take twice the nymber of gates as latches

2) so automatically delay is more for flipflops

3)power consumption is also more

What is slack?

The slack is the time delay difference from the expected delay(1/clock) to the actual delay in a particular path.

Slack may be +ve or -ve.

courtesy:***http://vlsi-guru.blogspot.com***
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Micro Processor Questions

No comments: Labels: ECE

- **What are the various registers in 8085?** - Accumulator register, Temporary register, Instruction register, Stack Pointer, Program Counter are the various registers in 8085 .
- **In 8085 name the 16 bit registers?** - Stack pointer and Program counter all have 16 bits.
- **What are the various flags used in 8085?** - Sign flag, Zero flag, Auxillary flag, Parity flag, Carry flag.
- **What is Stack Pointer?** - Stack pointer is a special purpose 16-bit register in the Microprocessor, which holds the address of the top of the stack.
- **What is Program counter?** - Program counter holds the address of either the first byte of the next instruction to be fetched for execution or the address of the next byte of a multi byte instruction, which has not been completely fetched. In both the cases it gets incremented automatically one by one as the instruction bytes get fetched. Also Program register keeps the address of the next instruction.
- **Which Stack is used in 8085?** - LIFO (Last In First Out) stack is used in 8085.In this type of Stack the last stored information can be retrieved first.
- **What happens when HLT instruction is executed in processor?** - The Micro Processor enters into Halt-State and the buses are tri-stated.
- **What is meant by a bus?** - A bus is a group of conducting lines that carriers data, address, & control signals.
- **What is Tri-state logic?** - Three Logic Levels are used and they are High, Low, High impedance state. The high and low are normal logic levels & high impedance state is electrical open circuit conditions. Tri-state logic has a third line called enable line.
- **Give an example of one address microprocessor?** - 8085 is a one address microprocessor.
- **In what way interrupts are classified in 8085?** - In 8085 the interrupts are classified as Hardware and Software interrupts.
- **What are Hardware interrupts?** - TRAP, RST7.5, RST6.5, RST5.5, INTR.
- **What are Software interrupts?** - RST0, RST1, RST2, RST3, RST4, RST5, RST6, RST7.
- **Which interrupt has the highest priority?** - TRAP has the highest priority.
- **Name 5 different addressing modes?** - Immediate, Direct, Register, Register indirect, Implied addressing modes.
- **How many interrupts are there in 8085?** - There are 12 interrupts in 8085.

- **What is clock frequency for 8085?** - 3 MHz is the maximum clock frequency for 8085.
- **What is the RST for the TRAP?** - RST 4.5 is called as TRAP.
- **In 8085 which is called as High order / Low order Register?** - Flag is called as Low order register & Accumulator is called as High order Register.
- **What are input & output devices?** - Keyboards, Floppy disk are the examples of input devices. Printer, LED / LCD display, CRT Monitor are the examples of output devices.
- **Can an RC circuit be used as clock source for 8085?** - Yes, it can be used, if an accurate clock frequency is not required. Also, the component cost is low compared to LC or Crystal.
- **Why crystal is a preferred clock source?** - Because of high stability, large Q (Quality Factor) & the frequency that doesn't drift with aging. Crystal is used as a clock source most of the times.
- **Which interrupt is not level-sensitive in 8085?** - RST 7.5 is a raising edge-triggering interrupt.
- **What does Quality factor mean?** - The Quality factor is also defined, as Q. So it is a number, which reflects the lossness of a circuit. Higher the Q, the lower are the losses.
- **What are level-triggering interrupt?** - RST 6.5 & RST 5.5 are level-triggering interrupts.

Links to this post

C Interview Questions For Hardware Engineer

No comments: Labels: C Interview Questions, ECE

1. What are the total number of lines written in C/C++? What is the most complicated/valuable program written in C/C++?
2. What compiler was used?
3. Have you studied buses? What types?
4. Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an instruction in a 5 stage machine? What is the throughput of this machine ?
5. How many bit combinations are there in a byte?
6. What is the difference between = and == in C?
7. Are you familiar with VHDL and/or Verilog?

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ECE Frequently Asked Interview Questions

14 comments: Labels: ECE

1. What is the difference between a latch and a flip flop. For the same input, how would the output look for a latch and for a flip-flop.

2. Finite state machines:

(2.1)Design a state-machine (or draw a state-diagram) to give an output '1' when the # of A's are even and # of B's are odd. The input is in the form of a serial-stream (one-bit per clock cycle). The inputs could be of the type A, B or C. At any given clock cycle, the output is a '1', provided the # of A's are even and # of B's are odd. At any given clock cycle, the output is a '0', if the above condition is not satisfied.

(2.2). To detect the sequence "abca" when the inputs can be a b c d.

3. minimize a boolean expression.

4. Draw transistor level nand gate.

5. Draw the cross-section of a CMOS inverter.

6. Deriving the vectors for the stuck at 0 and stuck at 1 faults.

7. Given a boolean expression he asked me to implement just with muxes but nothing else.

8. Draw Id Vds curves for mosfets and explain different regions.

9. Given the transfer characteristics of a black box draw the circuit for the black box.

10. Given a circuit and its inputs draw the outputs exact to the timing.

11. Given an inverter with a particular timing derive an inverter using the previous one but with the required timing other than the previous one.

12. Change the rise time and fall time of a given circuit by not changing the transistor sizes but by using current mirrors. 13. Some problems on clamping diodes.

These are some of the questions asked by Microsoft.
(I feel that these type of questions are asked even in Electrical Engineering interviews. Make sure you browse them.)

1. Given a rectangular (cuboidal for the puritans) cake with a rectangular piece removed (any size or orientation), how would you cut the remainder of the cake into two equal halves with one straight cut of a knife ?

2. You're given an array containing both positive and negative integers and required to find the sub-array with the largest sum (O(N) a la KBL).
Write a routine in C for the above.

3. Given an array of size N in which every number is between 1 and N, determine if there are any duplicates in it. You are allowed to destroy the array if you like.

4. Write a routine to draw a circle (x ** 2 + y ** 2 = r ** 2) without making use of any floating point computations at all.

5. Given only putchar (no sprintf, itoa, etc.) write a routine putlon the prints out an unsigned long in decimal.

6. Give a one-line C expression to test whether a number is a power of 2. [No loops allowed - it's a simple test.]

7. Given an array of characters which form a sentence of words, give an efficient algorithm to reverse the order of the words (no characters) in it.

8. How many points are there on the globe where by walking one mile south, one mile east and one mile north you reach the place where you started.

9. Give a very good method to count the number of ones in a 32 bit number. (caution: looping through testing each bit is not a solution)

10. What are the different ways to say, the value of x can be either a 0 or a 1. Apparently the if then else solution has a jump when written

out in assembly.
if (x == 0)
y=0
else
y =x

There is a logical, arithmetic and a datastructure soln to the above problem.

Logic design:

1. Draw the transistor level CMOS #input NAND or NOR gate.After drawing it lot of qestions on that ckt will be asked.

2. Transistor sizing for given rise time and fall time. How do you size it for equal rise and fall time.

3. Given a function whose inputs are dependent on its outputs. Design a sequential circuit.

4. Design a finite state machine to give a modulo 3 counter when x=0 and modulo 4 counter when x=1.

5. Given a boolean equation minimize it.

6. Given a boolean equation draw the transistor level minimum transistor circuit.

7. What is the function of a D-flipflop, whose inverted outputs are connected to its input ?

8. What will you do if you want to drive a large capacitance ?

Layout related questions:

1. asked me to layout the 3 input nand gate.

2. Later he asked me to modify it to consume as much less space as we can.

3. He also asked me about the transistor sizing.

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1. He asked me to draw the cross section of an inverter and asked me to show all the capacitances on it and reason for those capacitances.

2. Describe the latchup effect.

3. More about the tristate buffers.

3. What will be the voltage at the output node of a triostate buffer in its high impedence state. He gave a waveform for the input and asked me to draw the output waveform for that.

4. Posed a lot of questions on charge sharing problems and keeper circuits.

5. Asked me to draw the Id Vds curves for mosfet. Asked me to explain the regions and some couses for that curve like channel width modulation.

6. He asked me about the electron migration effect and methods to avoid it.

7. Asked me to draw the dynamic logic of a particular gate and then posed lots of tricky questions from the previous discussion.

8. He asked me to draw the 6 transistor contemporary sram cell and asked me to explain how the reading and writing is done in it.

9. Something about trip point.

Computer Architecture Questions:

1. Explain what is DMA?
2. what is pipelining?
3. what are superscalar machines and vliw machines?
4. what is cache?
5. what is cache coherency and how is it eliminated?
6. what is write back and write through caches?
7. what are different pipelining hazards and how are they eliminated.
8. what are different stages of a pipe?
9. eplain more about branch prediction in controlling the control hazards
10. Give examples of data hazards with pseudo codes.
11. Caluculating the number of sets given its way and size in a cache?
12. How is a block found in a cache?
13. scoreboard analysis.
14. What is miss penalty and give your own ideas to eliminate it.
15. How do you improve the cache performance.
16. Different addressing modes.
17. Computer arithmetic with two's complements.
18. About hardware and software interrupts.
19. What is bus contention and how do you eliminate it.
20. What is aliasing?
21) What is the difference between a latch and a flip flop?
22) What is the race around condition? How can it be overcome?
23) What is the purpose of cache? How is it used?
24) What are the types of memory management
Links to this post

Intel Interview Questions

1 comment: Labels: ECE

COMPUTER ARCHITECTURE QUESTIONS

1. For a single computer processor computer system, what is the purpose of a processor cache and describe its operation?

2. Explain the operation considering a two processor computer system with a cache for each processor.
What are the main issues associated with multiprocessor caches and how might you solve it?

3. Explain the difference between write through and write back cache.

4. Are you familiar with the term MESI?

5. Are you familiar with the term snooping?

STATE MACHINE QUESTIONS

1. Describe a finite state machine that will detect three consecutive coin tosses (of one coin) that results in heads.

2. In what cases do you need to double clock a signal before presenting it to a synchronous state machine?

SIGNAL LINE QUESTIONS

1. You have a driver that drives a long signal & connects to an input device. At the input device there is either overshoot,
undershoot or signal threshold violations, what can be done to correct this problem?

VALIDATION QUESTIONS:

What are the total number of lines written in C/C++? What is the most complicated/valuable program written in C/C++?

What compiler was used?

Have you studied busses? What types?

Have you studied pipelining? List the 5 stages of a 5 stage pipeline. Assuming 1 clock per stage, what is the latency of an instruction in a 5 stage machine? What is the throughput of this machine ?

How many bit combinations are there in a byte?

What is the difference between = and == in C?

Are you familiar with VHDL and/or Verilog?

MEMORY, I/O, CLOCK AND POWER QUESTIONS

1. What types of CMOS memories have you designed? What were their size? Speed? Configuration Process technology?

2. What work have you done on full chip Clock and Power distribution? What process technology and budgets were used?

3. What types of I/O have you designed? What were their size? Speed? Configuration? Voltage

requirements?

Process technology? What package was used and how did you model the package/system? What parasitic effects were considered?

4. What types of high speed CMOS circuits have you designed?

5. What transistor level design tools are you proficient with? What types of designs were they used on?

6. What products have you designed which have entered high volume production? What was your role in the silicon evaluation/product ramp? What tools did you use?

7. If not into production, how far did you follow the design and why did not you see it into production?
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ECE Interview Questions

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Digital Questions

Digital Electronics Questions

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Frequently Asked Interview Questions

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Job Hunting Guide for Analog/Mixed Signal Circuit Design

Electrical Engineering Technical Interview Questions/Review
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VLSI Design Interview Questions

No comments: Labels: ECE

VLSI Design Interview Questions

1. Explain why & how a MOSFET works
2. Draw Vds-Ids curve for a MOSFET. Now, show how this curve changes (a) with increasing Vgs (b) with increasing transistor width (c) considering Channel Length Modulation
3. Explain the various MOSFET Capacitances & their significance

4. Draw a CMOS Inverter. Explain its transfer characteristics
5. Explain sizing of the inverter
6. How do you size NMOS and PMOS transistors to increase the threshold voltage?
7. What is Noise Margin? Explain the procedure to determine Noise Margin
8. Give the expression for CMOS switching power dissipation
9. What is Body Effect?
10. Describe the various effects of scaling
11. Give the expression for calculating Delay in CMOS circuit
12. What happens to delay if you increase load capacitance?
13. What happens to delay if we include a resistance at the output of a CMOS circuit?
14. What are the limitations in increasing the power supply to reduce delay?
15. How does Resistance of the metal lines vary with increasing thickness and increasing length?
16. You have three adjacent parallel metal lines. Two out of phase signals pass through the outer two metal lines. Draw the waveforms in the center metal line due to interference. Now, draw the signals if the signals in outer metal lines are in phase with each other
17. What happens if we increase the number of contacts or via from one metal layer to the next?
18. Draw a transistor level two input NAND gate. Explain its sizing (a) considering Vth (b) for equal rise and fall times
19. Let A & B be two inputs of the NAND gate. Say signal A arrives at the NAND gate later than signal B. To optimize delay, of the two series NMOS inputs A & B, which one would you place near the output?
20. Draw the stick diagram of a NOR gate. Optimize it
21. For CMOS logic, give the various techniques you know to minimize power consumption
22. What is Charge Sharing? Explain the Charge Sharing problem while sampling data from a Bus
23. Why do we gradually increase the size of inverters in buffer design? Why not give the output of a circuit to one large inverter?
24. In the design of a large inverter, why do we prefer to connect small transistors in parallel (thus increasing effective width) rather than lay out one transistor with large width?
25. Given a layout, draw its transistor level circuit. (I was given a 3 input AND gate and a 2 input Multiplexer. You can expect any simple 2 or 3 input gates)
26. Give the logic expression for an AOI gate. Draw its transistor level equivalent. Draw its stick diagram
27. Why don't we use just one NMOS or PMOS transistor as a transmission gate?
28. For a NMOS transistor acting as a pass transistor, say the gate is connected to VDD, give the output for a square pulse input going from 0 to VDD
29. Draw a 6-T SRAM Cell and explain the Read and Write operations
30. Draw the Differential Sense Amplifier and explain its working. Any idea how to size this circuit? (Consider Channel Length Modulation)
31. What happens if we use an Inverter instead of the Differential Sense Amplifier?
32. Draw the SRAM Write Circuitry
33. Approximately, what were the sizes of your transistors in the SRAM cell? How did you arrive at those sizes?

34. How does the size of PMOS Pull Up transistors (for bit & bit- lines) affect SRAM's performance?
35. What's the critical path in a SRAM?
36. Draw the timing diagram for a SRAM Read. What happens if we delay the enabling of Clock signal?
37. Give a big picture of the entire SRAM Layout showing your placements of SRAM Cells, Row Decoders, Column Decoders, Read Circuit, Write Circuit and Buffers
38. In a SRAM layout, which metal layers would you prefer for Word Lines and Bit Lines? Why?
39. How can you model a SRAM at RTL Level?
40. What's the difference between Testing & Verification?
41. For an AND-OR implementation of a two input Mux, how do you test for Stuck-At-0 and Stuck-At-1 faults at the internal nodes? (You can expect a circuit with some redundant logic)
42. What is Latch Up? Explain Latch Up with cross section of a CMOS Inverter. How do you avoid Latch Up?

Infineon:

43. How do you tackle coupling when design deep submicron SRAM memories?

44 . Power Optimization Techniques for deep sub micron?

Digital Design Interview Questions

1. Give two ways of converting a two input NAND gate to an inverter
2. Given a circuit, draw its exact timing response. (I was given a Pseudo Random Signal Generator; you can expect any sequential ckt)
3. What are set up time & hold time constraints? What do they signify? Which one is critical for estimating maximum clock frequency of a circuit?
4. Give a circuit to divide frequency of clock cycle by two
5. Design a divide-by-3 sequential circuit with 50% duty circle. (Hint: Double the Clock)
6. Suppose you have a combinational circuit between two registers driven by a clock. What will you do if the delay of the combinational circuit is greater than your clock signal? (You can't resize the combinational circuit transistors)
7. The answer to the above question is breaking the combinational circuit and pipelining it. What will be affected if you do this?
8. What are the different Adder circuits you studied?
9. Give the truth table for a Half Adder. Give a gate level implementation of the same.
10. Draw a Transmission Gate-based D-Latch.
11. Design a Transmission Gate based XOR. Now, how do you convert it to XNOR? (Without inverting the output)
12. How do you detect if two 8-bit signals are same?
13. How do you detect a sequence of "1101" arriving serially from a signal line?
14. Design any FSM in VHDL or Verilog.

Links to this post

Micro Processor Interview Questions

2 comments: Labels: ECE

Micro Processor Questions

1. Which type of architecture 8085 has?
2. How many memory locations can be addressed by a microprocessor with 14 address lines?
3. 8085 is how many bit microprocessor?
4. Why is data bus bi-directional?
5. What is the function of accumulator?
6. What is flag, bus?
7. What are tri-state devices and why they are essential in a bus oriented system?
8. Why are program counter and stack pointer 16-bit registers?
9. What does it mean by embedded system?
10. What are the different addressing modes in 8085?
11. What is the difference between MOV and MVI?
12. What are the functions of RIM, SIM, IN?
13. What is the immediate addressing mode?
14. What are the different flags in 8085?
15. What happens during DMA transfer?
16. What do you mean by wait state? What is its need?
17. What is PSW?
18. What is ALE? Explain the functions of ALE in 8085.
19. What is a program counter? What is its use?
20. What is an interrupt?